FACULTE DES SCIENCES ET TECHNOLOGIES

Sciences Informatiques

Installer Ubuntu Serveur

TD N° 2 : ADMINISTRATION des Syestemes

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L3

Sous la direction du professeur :
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1. Objectif du TD

Ce TD a pour but de me familiariser avec les **sauvegardes automatisées sous Linux**, à la fois **locales** et **distantes**.

Les objectifs principaux sont :

- Comprendre le principe et l'importance des sauvegardes
- Mettre en place une **sauvegarde locale automatisée**
- Créer des sauvegardes datées et restaurables
- Automatiser les sauvegardes avec **cron**
- Mettre en place une **sauvegarde distante sécurisée** avec rsync et SSH
- Configurer une authentification par clé SSH pour des transferts sans mot de passe
- Vérifier et restaurer les fichiers sauvegardés

Exercice 1 : Sauvegardes locales

2.1 Contexte

Je suis administrateur système junior dans la startup **DevCloud**, une jeune entreprise qui développe des applications web.
Le serveur principal héberge :

- Les dépôts de code
- Les fichiers de configuration
- Les livrables clients

Suite à un incident (suppression accidentelle d'un dossier critique), la direction m'a demandé de mettre en place une **solution de sauvegarde locale automatisée**, fiable, simple et restaurable rapidement.

2.2 Architecture du serveur

- **Données à sauvegarder** : `/var/www/`, `/etc/`, `/home/devops/projects`

- **Destination des sauvegardes** : `/backup`

2.3 Démarche

Préparation des dossiers et fichiers

1- Création des dossiers

sudo mkdir /backup

mkdir -p /home/devops/projects

2- Création des fichiers de test

touch /home/devops/projects/fichier1.txt

touch /home/devops/projects/fichier2.txt

3- pour ajouter du contenu dedans

echo "TP sauvegarde" >> touch /home/devops/projects/fichier1.txt

echo "TP sauvegarde 1" >> touch /home/devops/projects/fichier2.txt

Cette étape prépare l’emplacement des sauvegardes et les dossiers de test.

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou executez : sudo pro status


Last login: Sat Mar 14 12:08:30 2026 from 10.200.196.171
blomy@blomyserveur:~$ sudo mkdir /backup
[sudo] password for blomy:
Sorry, try again.
[sudo] password for blomy:
blomy@blomyserveur:~$ mkdir -p /home/devops/projects
mkdir: cannot create directory ‘/home/devops’: Permission denied
blomy@blomyserveur:~$ sudo mkdir -p /home/devops/projects
blomy@blomyserveur:~$ cd ls
-bash: cd: ls: No such file or directory
blomy@blomyserveur:~$ ls
blomy@blomyserveur:~$ touch /home/devops/projects/fichier1.txt
touch: cannot touch '/home/devops/projects/fichier1.txt': Permission denied
blomy@blomyserveur:~$ sudo touch /home/devops/projects/fichier1.txt
blomy@blomyserveur:~$ sudo touch /home/devops/projects/fichier2.txt
blomy@blomyserveur:~$ sudo echo "TP sauvegarde" >> touch /home/devops/projects/fichier1.txt
blomy@blomyserveur:~$ sudo echo "TP sauvegarde 1" >> touch /home/devops/projects/fichier2.txt
blomy@blomyserveur:~$ ls
touch
blomy@blomyserveur:~$ cd status
-bash: cd: status: No such file or directory
blomy@blomyserveur:~$ ls -l
total 4
-rw-rw-r-- 1 blomy blomy 100 mars  14 16:38 touch
blomy@blomyserveur:~$ ls -l home/devops/projects
ls: cannot access 'home/devops/projects': No such file or directory
blomy@blomyserveur:~$ sudo ls -l home/devops/projects
ls: cannot access 'home/devops/projects': No such file or directory
blomy@blomyserveur:~$ sudo ls -l /home/devops/projects
total 0
-rw-r--r-- 1 root root 0 mars  14 16:34 fichier1.txt
-rw-r--r-- 1 root root 0 mars  14 16:34 fichier2.txt
blomy@blomyserveur:~$
```

Sauvegarde avec TAR

J’ai créé une archive compressée datée :

DATE=$(date +%F-%H-%M)

sudo tar -czf /backup/backup-$DATE.tar.gz /var/www /etc /home/devops/projects

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
blomy@blomyserveur:~$ ls
blomy@blomyserveur:~$ touch /home/devops/projects/fichier1.txt
touch: cannot touch '/home/devops/projects/fichier1.txt': Permission denied
blomy@blomyserveur:~$ sudo touch /home/devops/projects/fichier1.txt
blomy@blomyserveur:~$ sudo touch /home/devops/projects/fichier2.txt
blomy@blomyserveur:~$ sudo echo "TP sauvegarde" >> touch /home/devops/projects/fichier1.txt
blomy@blomyserveur:~$ sudo echo "TP sauvegarde 1" >> touch /home/devops/projects/fichier2.txt
blomy@blomyserveur:~$ ls
touch
blomy@blomyserveur:~$ cd status
-bash: cd: status: No such file or directory
blomy@blomyserveur:~$ ls -l
total 4
-rw-rw-r-- 1 blomy blomy 100 mars  14 16:38 touch
blomy@blomyserveur:~$ ls -l home/devops/projects
ls: cannot access 'home/devops/projects': No such file or directory
blomy@blomyserveur:~$ sudo ls -l home/devops/projects
ls: cannot access 'home/devops/projects': No such file or directory
blomy@blomyserveur:~$ sudo ls -l /home/devops/projects
total 0
-rw-r--r-- 1 root root 0 mars  14 16:34 fichier1.txt
-rw-r--r-- 1 root root 0 mars  14 16:34 fichier2.txt
blomy@blomyserveur:~$ DATE=$(date +%F-%H-%M)
blomy@blomyserveur:~$ sudo tar -czf /backup/backup-$DATE.tar.gz /var/www /etc /home/devops/projects
tar: Removing leading `/' from member names
tar: /var/www: Cannot stat: No such file or directory
tar: Removing leading `/' from hard link targets
tar: Exiting with failure status due to previous errors
blomy@blomyserveur:~$ ls /var
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp
blomy@blomyserveur:~$ sudo tar -czf /backup/backup-$DATE.tar.gz /home/devops/projects
tar: Removing leading `/' from member names
blomy@blomyserveur:~$ sudo mkdir -p /var/www
blomy@blomyserveur:~$ sudo tar -czf /backup/backup-$DATE.tar.gz /var/www /etc /home/devops/projects
tar: Removing leading `/' from member names
tar: Removing leading `/' from hard link targets
blomy@blomyserveur:~$
```

Vérification du contenu de l’archive :

tar -tzf /backup/backup-$DATE.tar.gz

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
blomy@blomyserveur:~$ sudo mkdir -p /var/www
blomy@blomyserveur:~$ sudo tar -czf /backup/backup-$DATE.tar.gz /var/www /etc /home/devops/projects
tar: Removing leading `/' from member names
tar: Removing leading `/' from hard link targets
blomy@blomyserveur:~$ tar -tzf /backup/backup-$DATE.tar.gz
var/www/
etc/
etc/bash.bashrc
etc/groff/
etc/groff/man.local
etc/groff/mdoc.local
etc/pki/
etc/pki/fwupd-metadata/
etc/pki/fwupd-metadata/GPG-KEY-Linux-Vendor-Firmware-Service
etc/pki/fwupd-metadata/LVFS-CA.pem
etc/pki/fwupd-metadata/GPG-KEY-Linux-Foundation-Metadata
etc/pki/fwupd/
etc/pki/fwupd/GPG-KEY-Linux-Foundation-Firmware
etc/pki/fwupd/GPG-KEY-Linux-Vendor-Firmware-Service
etc/pki/fwupd/LVFS-CA.pem
etc/sensors3.conf
etc/ifplugd/
etc/ifplugd/action.d/
etc/ifplugd/action.d/action_wpa
etc/libnl-3/
etc/libnl-3/pktloc
etc/libnl-3/classid
etc/sos/
etc/sos/cleaner/
etc/sos/presets.d/
etc/sos/sos.conf
etc/sos/extras.d/
etc/sos/groups.d/
etc/profile
etc/UPower/
etc/UPower/UPower.conf
etc/bash_completion.d/
etc/bash_completion.d/git-prompt
etc/hosts
```

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
etc/security/limits.d/
etc/security/namespace.d/
etc/security/time.conf
etc/update-motd.d/
etc/update-motd.d/50-motd-news
etc/update-motd.d/00-header
etc/update-motd.d/95-hwe-eol
etc/update-motd.d/98-fsck-at-reboot
etc/update-motd.d/50-landscape-sysinfo
etc/update-motd.d/10-help-text
etc/update-motd.d/91-contract-ua-esm-status
etc/update-motd.d/98-reboot-required
etc/update-motd.d/92-unattended-upgrades
etc/update-motd.d/91-release-upgrade
etc/update-motd.d/90-updates-available
etc/update-motd.d/85-fwupd
etc/update-motd.d/97-overlayroot
etc/sgml/
etc/sgml/catalog
etc/sgml/xml-core.cat
etc/protocols
etc/nsswitch.conf
etc/rc0.d/
etc/rc0.d/K01plymouth
etc/rc0.d/K01open-vm-tools
etc/rc0.d/K01unattended-upgrades
etc/rc0.d/K01cryptdisks-early
etc/rc0.d/K01uuidd
etc/rc0.d/K01cryptdisks
etc/rc0.d/K01open-iscsi
etc/rc0.d/K01iscsid
etc/usb_modeswitch.conf
etc/environment
home/devops/projects/
home/devops/projects/fichier1.txt
home/devops/projects/fichier2.txt
blomy@blomyserveur:~$
```

Tous les fichiers critiques sont présents dans l’archive.

Partie 3 : Sauvegarde incrémentale avec RSYNC

Installation de rsync :

sudo apt install rsync -y

Exécution d’une sauvegarde incrémentale :

rsync -av /var/www /etc /home/devops/projects /backup/

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
etc/update-motd.d/98-fsck-at-reboot
etc/update-motd.d/50-landscape-sysinfo
etc/update-motd.d/10-help-text
etc/update-motd.d/91-contract-ua-esm-status
etc/update-motd.d/98-reboot-required
etc/update-motd.d/92-unattended-upgrades
etc/update-motd.d/91-release-upgrade
etc/update-motd.d/90-updates-available
etc/update-motd.d/85-fwupd
etc/update-motd.d/97-overlayroot
etc/sgml/
etc/sgml/catalog
etc/sgml/xml-core.cat
etc/protocols
etc/nsswitch.conf
etc/rc0.d/
etc/rc0.d/K01plymouth
etc/rc0.d/K01open-vm-tools
etc/rc0.d/K01unattended-upgrades
etc/rc0.d/K01cryptdisks-early
etc/rc0.d/K01uuidd
etc/rc0.d/K01cryptdisks
etc/rc0.d/K01open-iscsi
etc/rc0.d/K01iscsid
etc/usb_modeswitch.conf
etc/environment
home/devops/projects/
home/devops/projects/fichier1.txt
home/devops/projects/fichier2.txt
blomy@blomyserveur:~$ sudo apt install rsync -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
rsync est déjà la version la plus récente (3.2.7-1ubuntu1.2).
rsync passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 5 non mis à jour.
blomy@blomyserveur:~$
```

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
etc/vmware-tools/vgauth/schemas/XMLSchema-hasFacetAndProperty.xsd
etc/vmware-tools/vgauth/schemas/XMLSchema-instance.xsd
etc/vmware-tools/vgauth/schemas/XMLSchema.dtd
etc/vmware-tools/vgauth/schemas/XMLSchema.xsd
etc/vmware-tools/vgauth/schemas/catalog.xml
etc/vmware-tools/vgauth/schemas/datatypes.dtd
etc/vmware-tools/vgauth/schemas/saml-schema-assertion-2.0.xsd
etc/vmware-tools/vgauth/schemas/xenc-schema.xsd
etc/vmware-tools/vgauth/schemas/xml.xsd
etc/vmware-tools/vgauth/schemas/xmldsig-core-schema.xsd
etc/wpa_supplicant/
etc/wpa_supplicant/action_wpa.sh
etc/wpa_supplicant/functions.sh
etc/wpa_supplicant/ifupdown.sh
etc/xdg/
etc/xdg/user-dirs.conf
etc/xdg/user-dirs.defaults
etc/xdg/autostart/
etc/xdg/autostart/snap-userd-autostart.desktop
etc/xdg/autostart/xdg-user-dirs.desktop
etc/xdg/systemd/
etc/xdg/systemd/user -> ../../systemd/user
etc/xml/
etc/xml/catalog
etc/xml/catalog.old
etc/xml/polkitd.xml
etc/xml/polkitd.xml.old
etc/xml/xml-core.xml
etc/xml/xml-core.xml.old
projects/
projects/fichier1.txt
projects/fichier2.txt
www/

sent 2.479.564 bytes  received 19.421 bytes  999.594,00 bytes/sec
total size is 2.395.352  speedup is 0,96
blomy@blomyserveur:~$
```

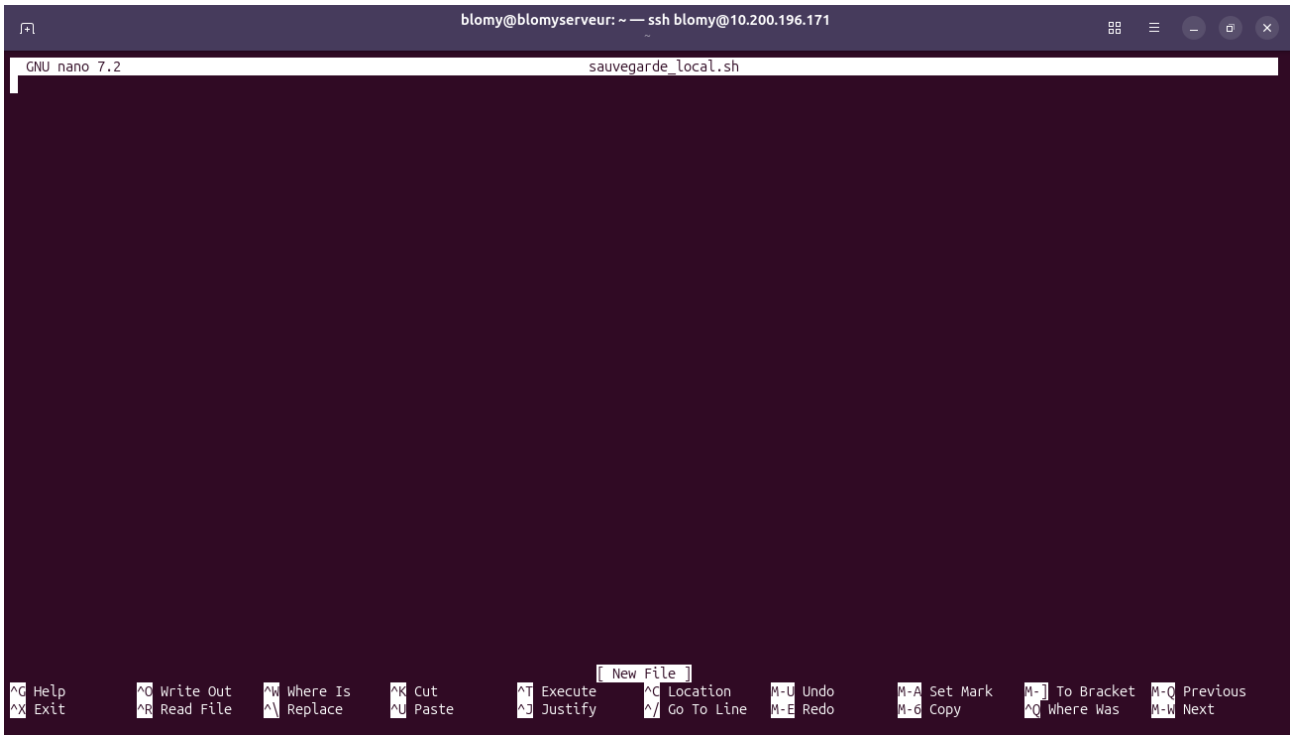
Seuls les fichiers modifiés sont copiés, ce qui optimise l’espace et le temps de transfert.

- Script de sauvegarde automatique

Création d’un script pour automatiser la sauvegarde :

nano sauvegarde_local.sh

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
total size is 2.394.962  speedup is 34,22
rsync error: some files/attrs were not transferred (see previous errors) (code 23) at main.c(1338) [sender=3.2.7]
blomy@blomyserveur:~$ sudo rsync -av /var/www /etc /home/devops/projects /backup/
sending incremental file list
etc/
etc/.pwd.lock
etc/.resolv.conf.systemd-resolved.bak
etc/.updated
etc/adduser.conf
etc/bash.bashrc
etc/bash_completion
etc/bindresvport.blacklist
etc/ca-certificates.conf
etc/crontab
etc/crypttab
etc/debconf.conf
etc/debian_version
etc/deluser.conf
etc/dhcpcd.conf
etc/e2scrub.conf
etc/environment
etc/ethertypes
etc/fstab
etc/fuse.conf
etc/gai.conf
etc/group
etc/group-
etc/gshadow
etc/gshadow-
etc/hdparm.conf
etc/host.conf
etc/hostname
etc/hosts
etc/hosts.allow
etc/hosts.deny
etc/inputrc
etc/issue
etc/issue.net
```



Contenu du script:

```
#!/bin/bash

DATE=$(date +%F-%H-%M)

tar -czf /backup/backup-$DATE.tar.gz /var/www /etc /home/devops/projects
```

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
  GNU nano 7.2                    sauvegarde_local.sh
#!/bin/bash
DATE=$(date +%F-%H-%M)
tar -czf /backup/backup-$DATE.tar.gz /var/www /etc /home/devops/projects

                                  [ Wrote 4 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut      ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket  M-Q Previous
^X Exit      ^R Read File  ^\ Replace    ^U Paste    ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy      ^Q Where Was    M-W Next
```

Rendre exécutable et tester :

chmod +x sauvegarde_local.sh

./sauvegarde_local.sh

ls /backup

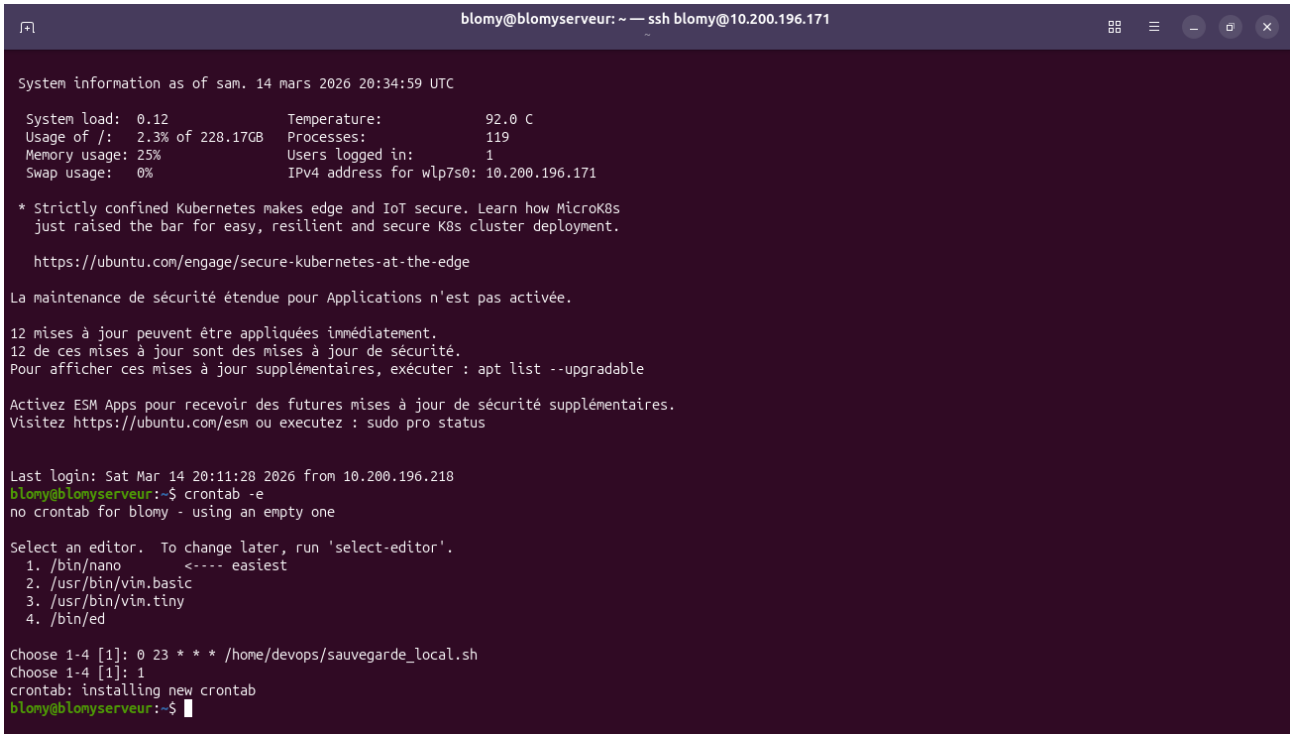
```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
: Permission denied
tar (child): Error is not recoverable: exiting now
tar: Removing leading `/' from hard link targets
tar: /backup/backup-2026-03-14-20-15.tar.gz: Cannot write: Broken pipe
tar: Child returned status 2
tar: Error is not recoverable: exiting now
backup-2026-03-14-17-04.tar.gz  etc  projects  www
blomy@blomyserveur:~$ ls /backup
backup-2026-03-14-17-04.tar.gz  etc  projects  www
blomy@blomyserveur:~$ sudo chmod +x sauvegarde_local.sh
[sudo] password for blomy:
Sorry, try again.
[sudo] password for blomy:
blomy@blomyserveur:~$ ls /backup
backup-2026-03-14-17-04.tar.gz  etc  projects  www
blomy@blomyserveur:~$ ./sauvegarde_local.sh
tar: Removing leading `/' from member names
tar (child): /backup/backup-2026-03-14-20-20.tar.gz: Cannot open: Permission denied
tar (child): Error is not recoverable: exiting now
tar: Removing leading `/' from hard link targets
tar: /backup/backup-2026-03-14-20-20.tar.gz: Cannot write: Broken pipe
tar: Child returned status 2
tar: Error is not recoverable: exiting now
blomy@blomyserveur:~$ sudo chmod +x sauvegarde_local.sh
blomy@blomyserveur:~$ ./sauvegarde_local.sh
tar: Removing leading `/' from member names
tar (child): /backup/backup-2026-03-14-20-21.tar.gz: Cannot open: Permission denied
tar (child): Error is not recoverable: exiting now
tar: Removing leading `/' from hard link targets
tar: /backup/backup-2026-03-14-20-21.tar.gz: Cannot write: Broken pipe
tar: Child returned status 2
tar: Error is not recoverable: exiting now
blomy@blomyserveur:~$ ls -ld /backup
drwxr-xr-x 5 root root 4096 mars  14 17:29 /backup
blomy@blomyserveur:~$ sudo ./sauvegarde_local.sh
tar: Removing leading `/' from member names
tar: Removing leading `/' from hard link targets
blomy@blomyserveur:~$ sudo sauvegarde_local.sh
```

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171
Sorry, try again.
[sudo] password for blomy:
blomy@blomyserveur:~$ ls /backup
backup-2026-03-14-17-04.tar.gz  etc  projects  www
blomy@blomyserveur:~$ ./sauvegarde_local.sh
tar: Removing leading `/' from member names
tar (child): /backup/backup-2026-03-14-20-20.tar.gz: Cannot open: Permission denied
tar (child): Error is not recoverable: exiting now
tar: Removing leading `/' from hard link targets
tar: /backup/backup-2026-03-14-20-20.tar.gz: Cannot write: Broken pipe
tar: Child returned status 2
tar: Error is not recoverable: exiting now
blomy@blomyserveur:~$ sudo chmod +x sauvegarde_local.sh
blomy@blomyserveur:~$ ./sauvegarde_local.sh
tar: Removing leading `/' from member names
tar (child): /backup/backup-2026-03-14-20-21.tar.gz: Cannot open: Permission denied
tar (child): Error is not recoverable: exiting now
tar: Removing leading `/' from hard link targets
tar: /backup/backup-2026-03-14-20-21.tar.gz: Cannot write: Broken pipe
tar: Child returned status 2
tar: Error is not recoverable: exiting now
blomy@blomyserveur:~$ ls -ld /backup
drwxr-xr-x 5 root root 4096 mars  14 17:29 /backup
blomy@blomyserveur:~$ sudo ./sauvegarde_local.sh
tar: Removing leading `/' from member names
tar: Removing leading `/' from hard link targets
blomy@blomyserveur:~$ sudo sauvegarde_local.sh
sudo: sauvegarde_local.sh: command not found
blomy@blomyserveur:~$ sudo .sauvegarde_local.sh
sudo: .sauvegarde_local.sh: command not found
blomy@blomyserveur:~$ sudo/.sauvegarde_local.sh
-bash: sudo/.sauvegarde_local.sh: No such file or directory
blomy@blomyserveur:~$ sudo./sauvegarde_local.sh
-bash: sudo./sauvegarde_local.sh: No such file or directory
blomy@blomyserveur:~$ ls /backup
backup-2026-03-14-17-04.tar.gz  backup-2026-03-14-20-24.tar.gz  etc  projects  www
blomy@blomyserveur:~$
```

Partie 5 : Automatisation avec CRON

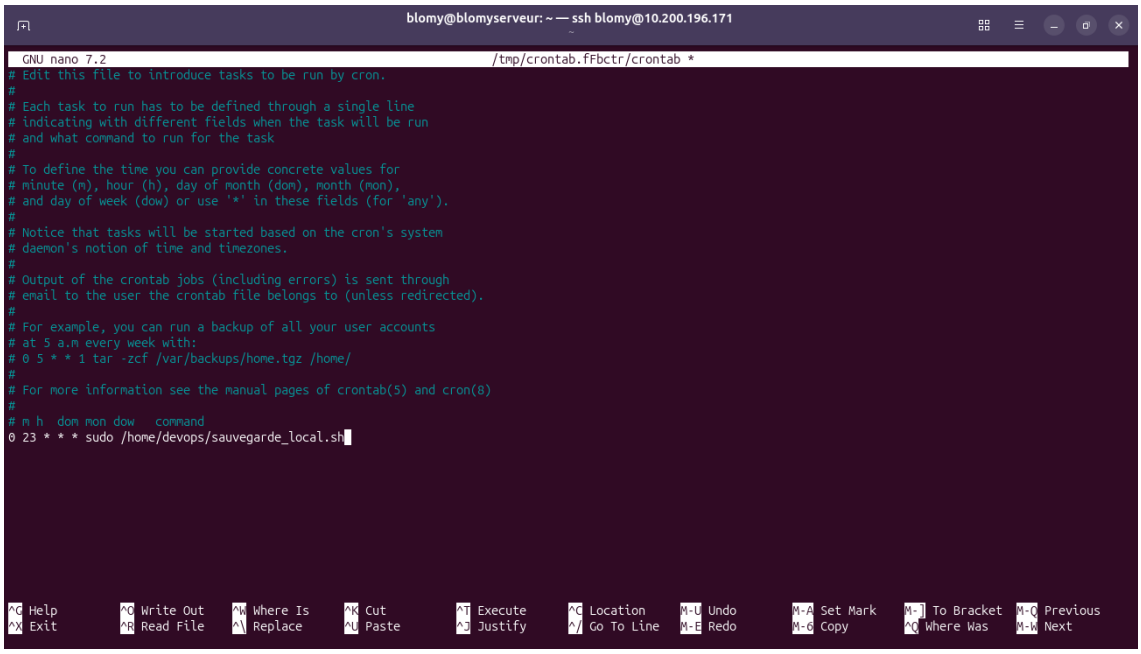
Pour automatiser le script toutes les nuits à 23h :

crontab -e



Ajouter la ligne :

0 23 * * * /home/devops/sauvegarde_local.sh



Cette configuration permet une sauvegarde quotidienne automatique.

Restauration des fichiers

Suppression d’un dossier pour tester la restauration :

rm -r /home/devops/projects

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171

   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

La maintenance de sécurité étendue pour Applications n'est pas activée.

12 mises à jour peuvent être appliquées immédiatement.
12 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou executez : sudo pro status


Last login: Sat Mar 14 20:11:28 2026 from 10.200.196.218
blomy@blomyserveur:~$ crontab -e
no crontab for blomy - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny
  4. /bin/ed

Choose 1-4 [1]: 0 23 * * * /home/devops/sauvegarde_local.sh
Choose 1-4 [1]: 1
crontab: installing new crontab
blomy@blomyserveur:~$ ls /backup
backup-2026-03-14-17-04.tar.gz  backup-2026-03-14-20-24.tar.gz  etc  projects  www
blomy@blomyserveur:~$ rm -r /home/devops/projects
rm: descend into write-protected directory '/home/devops/projects'? yes
rm: remove write-protected regular empty file '/home/devops/projects/fichier1.txt'? yes
rm: cannot remove '/home/devops/projects/fichier1.txt': Permission denied
rm: remove write-protected regular empty file '/home/devops/projects/fichier2.txt'? yes
rm: cannot remove '/home/devops/projects/fichier2.txt': Permission denied
blomy@blomyserveur:~$ sudo rm -r /home/devops/projects
[sudo] password for blomy:
blomy@blomyserveur:~$ crontab -1
```

Restauration depuis l’archive la plus récente :

sudo tar -xzf /backup/backup-2026-03-14-17-04.tar.gz -C /

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171

blomy@blomyserveur:~$ sudo rm -r /home/devops/projects
rm: cannot remove '/home/devops/projects': No such file or directory
blomy@blomyserveur:~$ tar -xzf /backup/backup-AAAA-MM-JJ-HH-MM.tar.gz -C /
tar (child): /backup/backup-AAAA-MM-JJ-HH-MM.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
blomy@blomyserveur:~$ ls /home/devops
blomy@blomyserveur:~$ sudo tar -xzf /backup/backup-2026-03-14-17-04.tar.gz -C /
blomy@blomyserveur:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 23 * * * sudo /home/devops/sauvegarde_local.sh
blomy@blomyserveur:~$ ls /backup
backup-2026-03-14-17-04.tar.gz  backup-2026-03-14-20-24.tar.gz  etc  projects  www
blomy@blomyserveur:~$
```

Les fichiers sont restaurés correctement.

Exercice 2 : Sauvegarde distante RSYNC +SSH

3.1 Contexte professionnel

La startup DevCloud possède :

- **SRV-PROD** : serveur principal (applications, code, configs)
- **SRV-BACKUP** : serveur interne dédié aux sauvegardes

Suite à une panne disque, la direction impose :

- Une **sauvegarde automatique distante**
- Sans saisie de mot de passe
- Avec transfert incrémental optimisé
- Avec logs pour audit

3.2 Architecture

- **Serveur source (SRV-PROD)** : `/var/www/`, `/etc/`, `/home/devops/projects`
- **Serveur de sauvegarde (SRV-BACKUP)** : `/backup/devcloud`
- **Utilisateur dédié** : `backupuser`

3.3 DémarcheS

PARTIE 1 : Préparation du serveur DE SAUVEGARDE

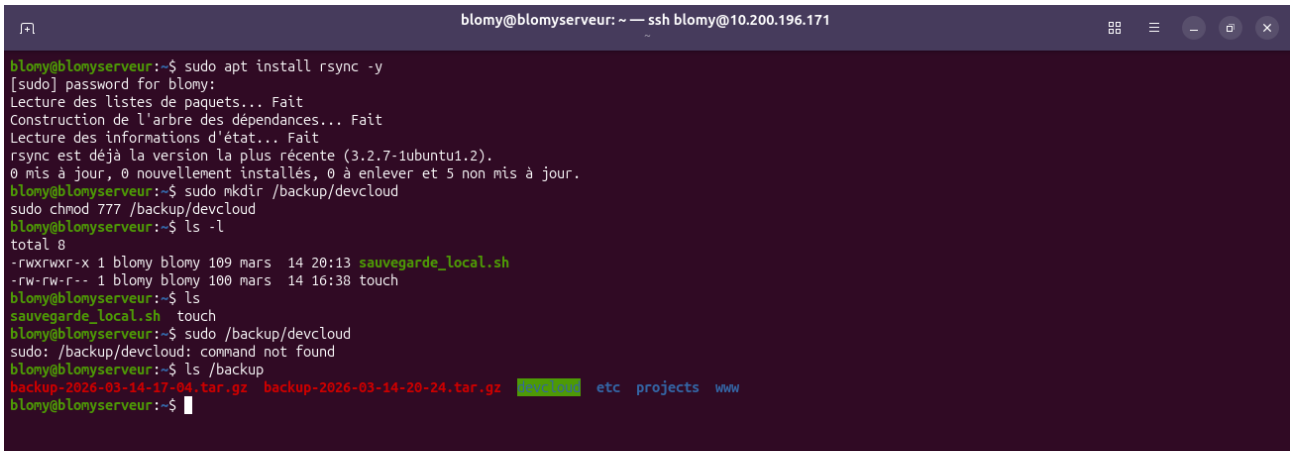
1- J'INSTALLE RSYNC sur le serveur distant

sudo apt install rsync -y

2- Je crée des dossiers de sauvegarde

sudo mkdir /backup/devcloud

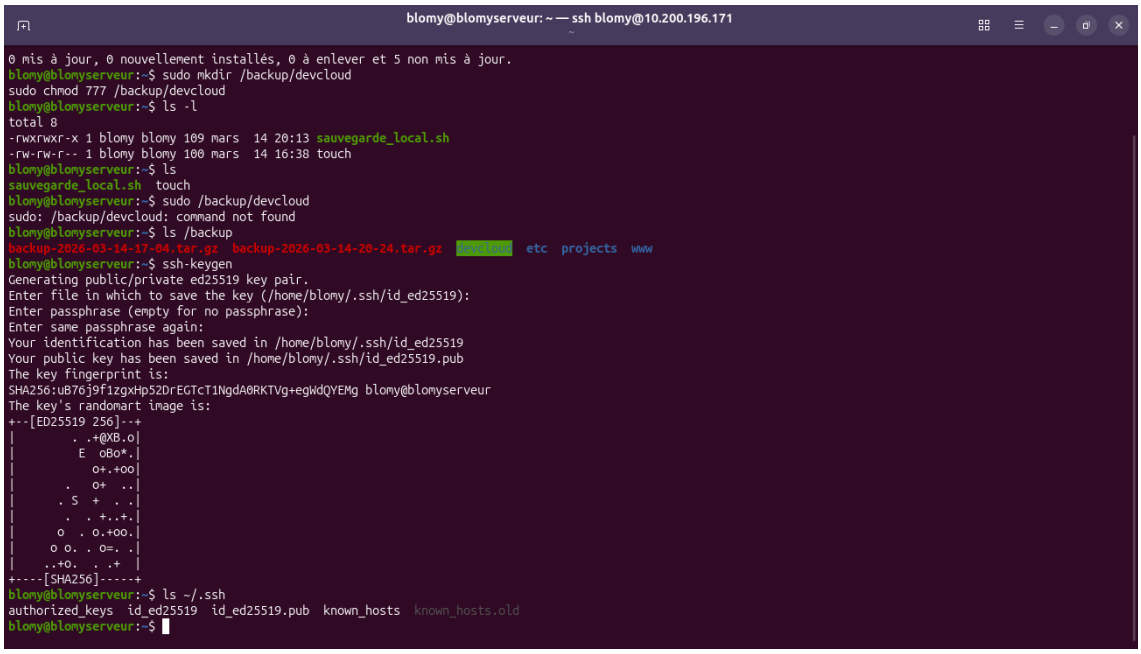
sudo chmod 777 /backup/devcloud



PARTIE 2 : Authentification SSH par clé

1- Sur la machine cliente je fais:

ssh-keygen



2- copie la cle vers le serveur

ssh-copy-id backupuser@SRV-BACKUP

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171

blomy@blomyserveur:~$ ssh-copy-id backupuser@10.200.196.171
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/blomy/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
backupuser@10.200.196.171's password:
Permission denied, please try again.
backupuser@10.200.196.171's password:
Permission denied, please try again.
backupuser@10.200.196.171's password:
backupuser@10.200.196.171: Permission denied (publickey,password).
blomy@blomyserveur:~$
blomy@blomyserveur:~$ sudo adduser backupuser
sudo passwd backupuser
[sudo] password for blomy:
info: Adding user `backupuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `backupuser' (1001) ...
info: Adding new user `backupuser' (1001) with group `backupuser (1001)' ...
info: Creating home directory `/home/backupuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for backupuser
Enter the new value, or press ENTER for the default
        Full Name []: blomay
        Room Number []: 2
        Work Phone []: 40036216
        Home Phone []: 37354271
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `backupuser' to supplemental / extra groups `users' ...
info: Adding user `backupuser' to group `users' ...
New password:
Retype new password:
passwd: password updated successfully
blomy@blomyserveur:~$
```

```
blomy@blomyserveur: ~ — ssh blomy@10.200.196.171

blomy@blomyserveur:~$ sudo adduser backupuser
sudo passwd backupuser
[sudo] password for blomy:
info: Adding user `backupuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `backupuser' (1001) ...
info: Adding new user `backupuser' (1001) with group `backupuser (1001)' ...
info: Creating home directory `/home/backupuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for backupuser
Enter the new value, or press ENTER for the default
        Full Name []: blomay
        Room Number []: 2
        Work Phone []: 40036216
        Home Phone []: 37354271
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `backupuser' to supplemental / extra groups `users' ...
info: Adding user `backupuser' to group `users' ...
New password:
Retype new password:
passwd: password updated successfully
blomy@blomyserveur:~$ ssh-copy-id backupuser@10.200.196.171
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/blomy/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
backupuser@10.200.196.171's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'backupuser@10.200.196.171'"
and check to make sure that only the key(s) you wanted were added.

blomy@blomyserveur:~$
```

3- vérification de la connexion

ssh backupuser@10.200.196.171

```
backupuser@blomyserveur: ~ — ssh blomy@10.200.196.171

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'backupuser@10.200.196.171'"
and check to make sure that only the key(s) you wanted were added.

blomy@blomyserveur:~$ ssh backupuser@SRV-BACKUP
ssh: Could not resolve hostname srv-backup: Temporary failure in name resolution
blomy@blomyserveur:~$ ssh backupuser@10.200.196.171
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of sam. 14 mars 2026 21:56:27 UTC

  System load:  0.09               Temperature:             92.0 C
  Usage of /:   2.3% of 228.17GB   Processes:               124
  Memory usage: 25%                Users logged in:         1
  Swap usage:   0%                 IPv4 address for wlp7s0: 10.200.196.171

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

La maintenance de sécurité étendue pour Applications n'est pas activée.

12 mises à jour peuvent être appliquées immédiatement.
12 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou executez : sudo pro status


backupuser@blomyserveur:~$
```

La connexion fonctionne sans mot de passe.

PARTIE 3 : PREMIER Sauvegarde distante avec RSYNC

1) sur la machine cliente j'ai exécuté cette commande :

rsync -avz /var/www /etc /home/devops/projects backupuser@10.236.174.171:/backup/devcloud

```
backupuser@blomyserveur: ~ — ssh blomy@10.236.174.171

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou executez : sudo pro status


Last login: Sat Mar 14 20:35:02 2026 from 10.200.196.218
blomy@blomyserveur:~$ rsync -avz /var/www /etc /home/devops/projects backupuser@10.236.174.171:/backup/devcloud
The authenticity of host '10.236.174.171 (10.236.174.171)' can't be established.
ED25519 key fingerprint is SHA256:wktqHpD18lkA91UpK8rgC6pVudB/B2NfhvhhWVuBFq8.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.236.174.171' (ED25519) to the list of known hosts.
sending incremental file list
rsync: [sender] opendir "/etc/credstore.encrypted" failed: Permission denied (13)
rsync: [sender] opendir "/etc/credstore" failed: Permission denied (13)
rsync: [sender] opendir "/etc/multipath" failed: Permission denied (13)
rsync: [sender] opendir "/etc/polkit-1/rules.d" failed: Permission denied (13)
etc/
rsync: [sender] send_files failed to open "/etc/.pwd.lock": Permission denied (13)
etc/.resolv.conf.systemd-resolved.bak
etc/.updated
etc/adduser.conf
etc/bash.bashrc
etc/bash_completion
etc/bindresvport.blacklist
etc/ca-certificates.conf
etc/crontab
etc/crypttab
etc/debconf.conf
etc/debian_version
etc/deluser.conf
etc/dhcpcd.conf
etc/e2scrub.conf
etc/environment
etc/ethertypes
etc/fstab
etc/fuse.conf
```

```
backupuser@blomyserveur: ~ — ssh blomy@10.236.174.171

etc/vmware-tools/vgauth/schemas/
etc/vmware-tools/vgauth/schemas/XMLSchema-hasFacetAndProperty.xsd
etc/vmware-tools/vgauth/schemas/XMLSchema-instance.xsd
etc/vmware-tools/vgauth/schemas/XMLSchema.dtd
etc/vmware-tools/vgauth/schemas/XMLSchema.xsd
etc/vmware-tools/vgauth/schemas/catalog.xml
etc/vmware-tools/vgauth/schemas/datatypes.dtd
etc/vmware-tools/vgauth/schemas/saml-schema-assertion-2.0.xsd
etc/vmware-tools/vgauth/schemas/xenc-schema.xsd
etc/vmware-tools/vgauth/schemas/xml.xsd
etc/vmware-tools/vgauth/schemas/xmldsig-core-schema.xsd
etc/wpa_supplicant/
etc/wpa_supplicant/action_wpa.sh
etc/wpa_supplicant/functions.sh
etc/wpa_supplicant/ifupdown.sh
etc/xdg/
etc/xdg/user-dirs.conf
etc/xdg/user-dirs.defaults
etc/xdg/autostart/
etc/xdg/autostart/snap-userd-autostart.desktop
etc/xdg/autostart/xdg-user-dirs.desktop
etc/xdg/systemd/
etc/xdg/systemd/user -> ../../systemd/user
etc/xml/
etc/xml/catalog
etc/xml/catalog.old
etc/xml/polkitd.xml
etc/xml/polkitd.xml.old
etc/xml/xml-core.xml
etc/xml/xml-core.xml.old
projects/
projects/fichier1.txt
projects/fichier2.txt
www/

sent 641.519 bytes  received 19.319 bytes  23.187,30 bytes/sec
total size is 2.395.439  speedup is 3,62
rsync error: some files/attrs were not transferred (see previous errors) (code 23) at main.c(1338) [sender=3.2.7]
```

J'ai connecte sur le serveur de sauvegade pour verifier si les fichier ont ete envoye avec la commande : ssh backupuser@10.236.174.171/ puis je fais :

ls /backup/devcloud pour vérifier su cote de serveur distant.

```
backupuser@blomyserveur: ~ — ssh blomy@10.236.174.171

total size is 2.395.439  speedup is 3,62
rsync error: some files/attrs were not transferred (see previous errors) (code 23) at main.c(1338) [sender=3.2.7]
blomy@blomyserveur:~$ ls /backup/devcloud
etc  projects  www
blomy@blomyserveur:~$ ssh backupuser@10.236.174.171
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of dim. 15 mars 2026 10:34:46 UTC

  System load:  1.48               Temperature:             81.0 C
  Usage of /:   2.4% of 228.17GB   Processes:               134
  Memory usage: 41%                Users logged in:         1
  Swap usage:   0%                 IPv4 address for wlp7s0: 10.236.174.171

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

La maintenance de sécurité étendue pour Applications n'est pas activée.

12 mises à jour peuvent être appliquées immédiatement.
12 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou executez : sudo pro status


Last login: Sat Mar 14 21:56:33 2026 from 10.200.196.171
backupuser@blomyserveur:~$ ls /backup/devcloud
etc  projects  www
backupuser@blomyserveur:~$
```

Figure 1: La commande `rsync` a correctement synchronisé les fichiers entre :

- ***la machine cliente et le serveur de sauvegarde distant via SSH***

*La présence des dossiers `etc`, `projects` et `www` sur le serveur distant confirme que **la sauvegarde distante est fonctionnelle**.*

PARTIE 5 : Script de sauvegarde distante

1) Sur la machine cliente :

nano sauvegarde_distante.sh
Contenu du script:

```
#!/bin/bash

SOURCE="/var/www /etc /home/devops/projects"

DEST="backupuser@SRV-BACKUP:/backup/devcloud"

LOG="/home/devops/backup.log"

rsync -avz $SOURCE $DEST >> $LOG 2>&1
```

2) Rendre le script exécutable et verifier

chmod +x sauvegarde_distante.sh et **./sauvegarde_distante.sh**

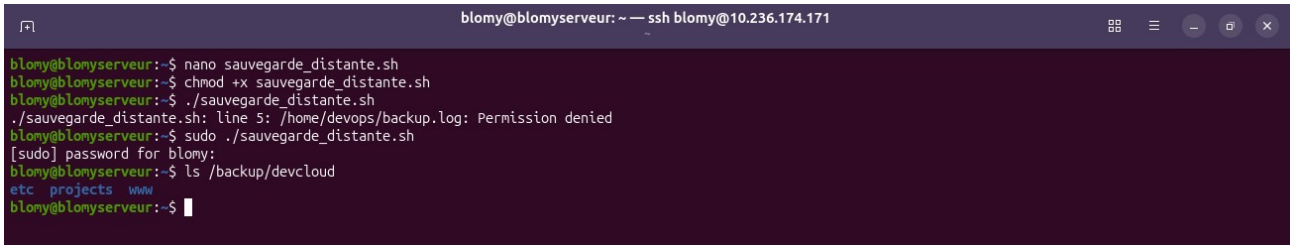
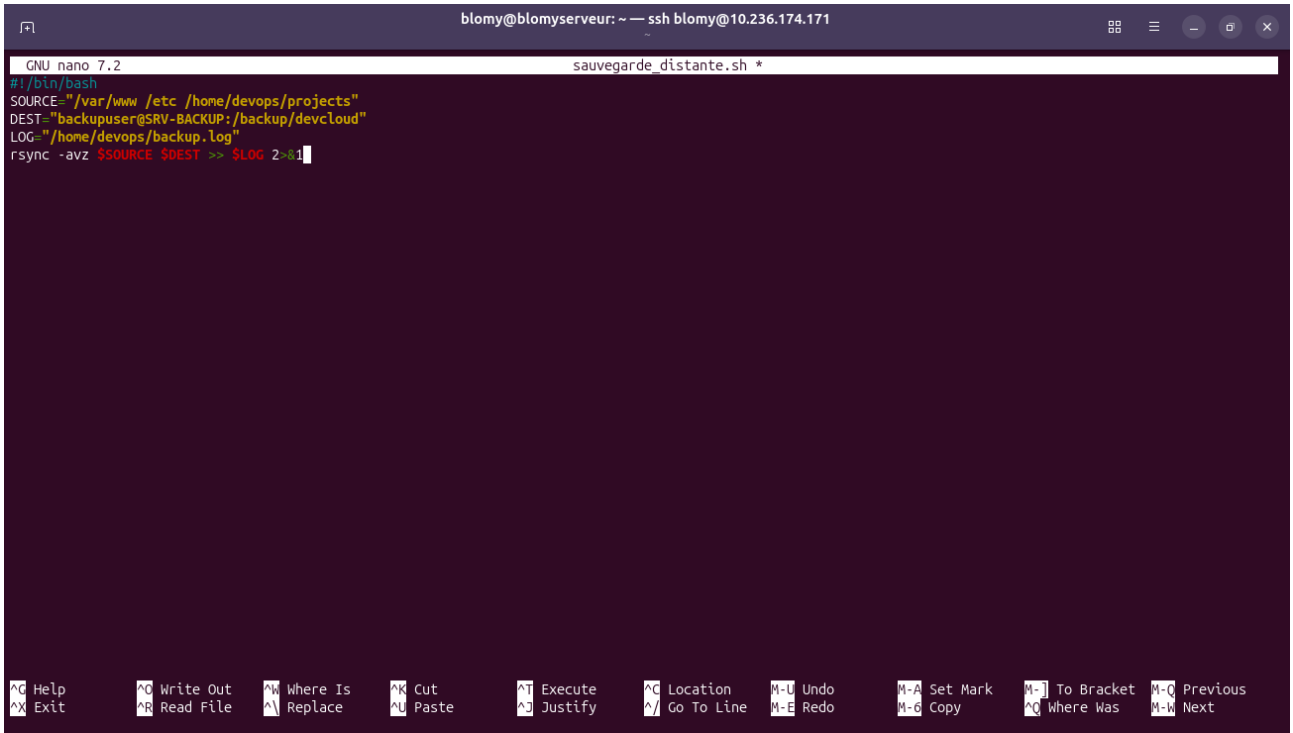


Figure 2: La sauvegarde a été faite avec succès



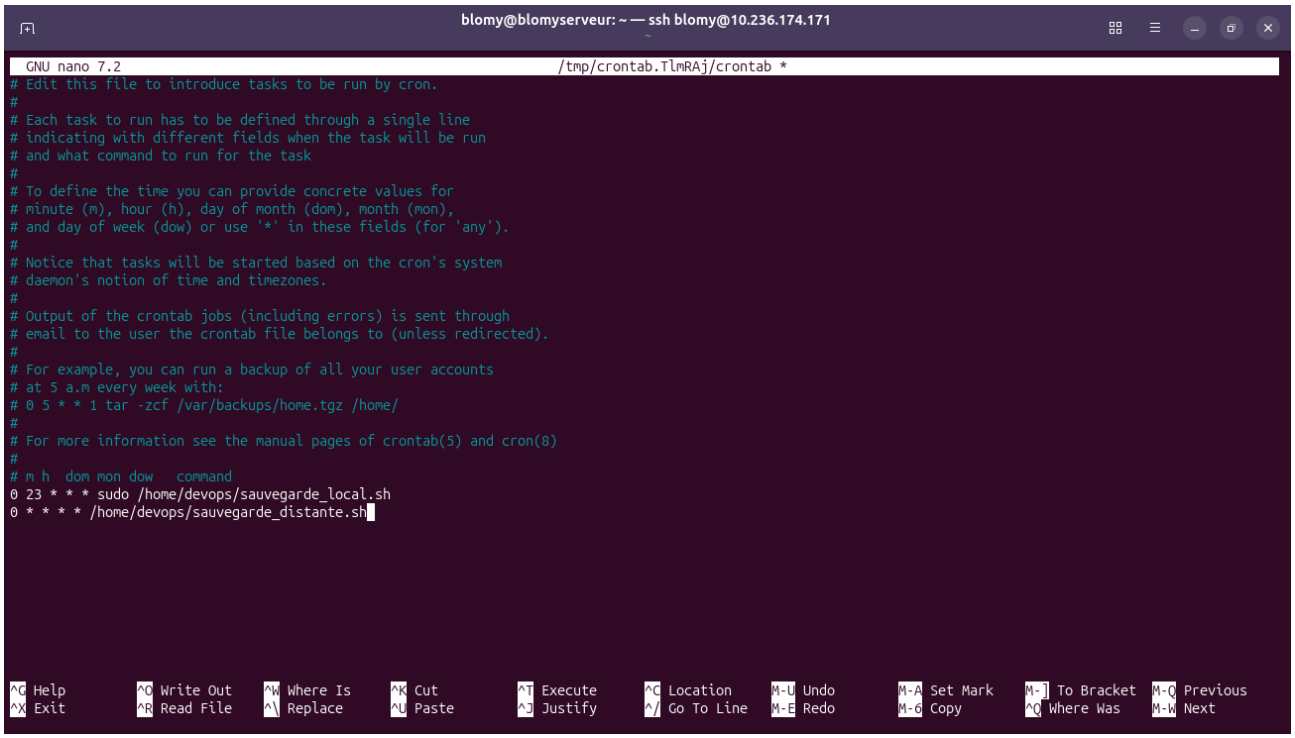
Partie 6 : Automatisation avec CRON

Exécution automatique toutes les heures :

crontab -e

Ajouter :

0 * * * * /home/devops/sauvegarde_distante.sh



Je verifie avec la commande : crontab -l

```
blomy@blomyserveur: ~ — ssh blomy@10.236.174.171

blomy@blomyserveur:~$ crontab -e
crontab: installing new crontab
blomy@blomyserveur:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 23 * * * sudo /home/devops/sauvegarde_local.sh
0 * * * * /home/devops/sauvegarde_distante.sh
blomy@blomyserveur:~$
```

Partie 7 : Restauration depuis le serveur distant

1) j'affiche le contenu du dossier : ls /home/devops/projects

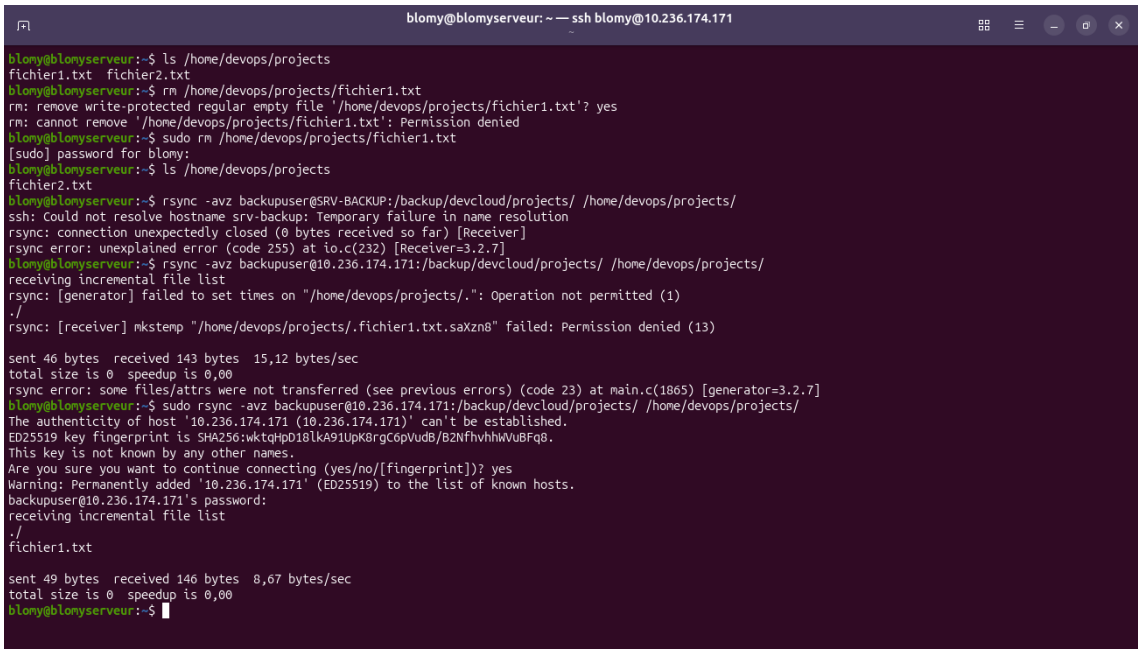
```
blomy@blomyserveur: ~ — ssh blomy@10.236.174.171

blomy@blomyserveur:~$ crontab -e
crontab: installing new crontab
blomy@blomyserveur:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 23 * * * sudo /home/devops/sauvegarde_local.sh
0 * * * * /home/devops/sauvegarde_distante.sh
blomy@blomyserveur:~$ ls /home/devops/projects
fichier1.txt  fichier2.txt
blomy@blomyserveur:~$
```

2) Suppression d'un fichier pour test :

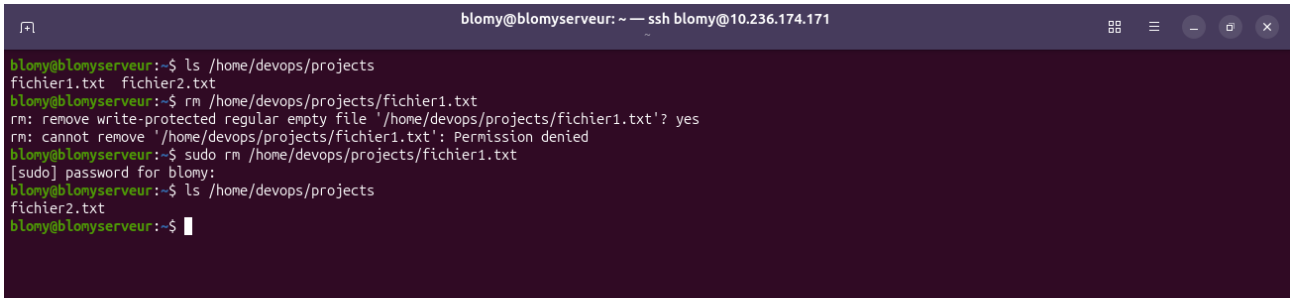
Sudo rm /home/devops/projects/fichier1.txt

3) Restauration du fichier supprime:

sudo rsync -avz backupuser@10.236.174.171:/backup/devcloud/projects/ /home/devops/projects/



4) je verifie si le fichier supprime est revenu a sa place primordiale



```
blomy@blomyserveur: ~ — ssh blomy@10.236.174.171

fichier1.txt  fichier2.txt
blomy@blomyserveur:~$ rm /home/devops/projects/fichier1.txt
rm: remove write-protected regular empty file '/home/devops/projects/fichier1.txt'? yes
rm: cannot remove '/home/devops/projects/fichier1.txt': Permission denied
blomy@blomyserveur:~$ sudo rm /home/devops/projects/fichier1.txt
[sudo] password for blomy:
blomy@blomyserveur:~$ ls /home/devops/projects
fichier2.txt
blomy@blomyserveur:~$ rsync -avz backupuser@SRV-BACKUP:/backup/devcloud/projects/ /home/devops/projects/
ssh: Could not resolve hostname srv-backup: Temporary failure in name resolution
rsync: connection unexpectedly closed (0 bytes received so far) [Receiver]
rsync error: unexplained error (code 255) at io.c(232) [Receiver=3.2.7]
blomy@blomyserveur:~$ rsync -avz backupuser@10.236.174.171:/backup/devcloud/projects/ /home/devops/projects/
receiving incremental file list
rsync: [generator] failed to set times on "/home/devops/projects/.": Operation not permitted (1)
./
rsync: [receiver] mkstemp "/home/devops/projects/.fichier1.txt.saXzn8" failed: Permission denied (13)

sent 46 bytes  received 143 bytes  15,12 bytes/sec
total size is 0  speedup is 0,00
rsync error: some files/attrs were not transferred (see previous errors) (code 23) at main.c(1865) [generator=3.2.7]
blomy@blomyserveur:~$ sudo rsync -avz backupuser@10.236.174.171:/backup/devcloud/projects/ /home/devops/projects/
The authenticity of host '10.236.174.171 (10.236.174.171)' can't be established.
ED25519 key fingerprint is SHA256:wktqHpD18lkA91UpK8rgC6pVudB/B2NfhvhhWVuBFq8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.236.174.171' (ED25519) to the list of known hosts.
backupuser@10.236.174.171's password:
receiving incremental file list
./
fichier1.txt

sent 49 bytes  received 146 bytes  8,67 bytes/sec
total size is 0  speedup is 0,00
blomy@blomyserveur:~$ ls /home/devops/projects
fichier1.txt  fichier2.txt
blomy@blomyserveur:~$
```

Le fichier supprimé est revenu correctement.

3. Résultats et observations

- Les sauvegardes locales avec **tar** sont simples et compressées.
- **Rsync** permet des sauvegardes incrémentales rapides et efficaces.
- **Cron** automatise parfaitement la sauvegarde régulière.
- La sauvegarde distante sécurisée avec SSH évite la saisie de mot de passe et protège les données.
- La restauration est fiable et rapide.

4. Réponses aux questions

1. **Différence entre tar et rsync** :
    - tar : crée une archive complète compressée.
    - rsync : synchronise les fichiers de manière incrémentale.
2. **Différence entre scp et rsync** :
    - scp : copie simple, tout est recopié.
    - rsync : copie intelligente, seuls les fichiers modifiés sont transférés.
3. **Pourquoi automatiser les sauvegardes** :
    - Éviter les pertes de données
    - Gain de temps
    - Sauvegarde régulière et fiable
    - Sécurité et continuité du service
4. **Où sont stockées les tâches cron** :
    - /var/spool/cron et consultables avec crontab -l
5. **Planifier une sauvegarde tous les jours à 23h** :

0 23 * * * /chemin/vers/script.sh

5. Conclusion

Ce TD m'a permis de comprendre **l'importance des sauvegardes** et comment les mettre en œuvre sur Linux. J'ai appris à :

- Créer des archives compressées avec **tar**

- Synchroniser des fichiers avec **rsync**
- Automatiser les sauvegardes avec **cron**
- Mettre en place une sauvegarde distante sécurisée via **SSH**
- Restaurer rapidement des données locales ou distantes

Difficultés rencontrées :

- Syntaxe de cron
- Configuration SSH

Solutions :

- Vérification pas à pas de chaque ligne de cron
- Génération et copie de la clé SSH correctement

Le TD est réussi et les objectifs sont atteints.

6. Annexes

- Captures d’écran des commandes et résultats
- Scripts sauvegarde.sh et sauvegarde_distante.sh
- Logs des sauvegardes distantes